LatticeQuant: Lattice Quantization Principles for KV Cache Compression

Jeehyun Yoon
Independent Researcher

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Abstract

We establish a general lattice-based quantization framework for KV cache compression and derive two complementary theoretical results. **Theorem 1** proves that any scalar fixed-length quantization scheme suffers an asymptotic Gaussian-normalized distortion gap of at least $\pi\sqrt{3}/2 \approx 2.72$, decomposed into a scalar-lattice cell penalty ($\pi e/6 \approx 1.424$) and a fixed-length rate penalty ($3\sqrt{3}/e \approx 1.911$). **Theorem 2** establishes a *general lattice achievability principle*: for any n-dimensional lattice Λ with normalized second moment $G(\Lambda)$, entropy-coded nearest-neighbor lattice quantization achieves asymptotically a gap of at most $2\pi e\, G(\Lambda)$ under a broad class of continuous sources. As the strongest practical instantiation, E_8 lattice quantization with parity-aware entropy coding achieves a gap of at most 1.224, a 2.22× reduction in the Gaussian-normalized distortion constant relative to the scalar fixed-length barrier. Experiments on Llama-3.2-1B and Llama-3.1-8B confirm that the theoretical predictions hold on real KV caches (gap deviation $< 0.4\%$). On Llama-3.1-8B, 4-bit quantization—corresponding to a 4× theoretical KV-storage reduction—incurs only +1.94% perplexity increase on wikitext2, and 5-bit quantization is near-lossless at +0.51%.

1 Introduction

Efficient KV cache quantization is essential for deploying large language models under memory constraints. TurboQuant (Ashkboos et al., ICLR 2026) applies a randomized Hadamard transform (RHT) to produce approximately Gaussian coordinates, then quantizes each coordinate independently using a scalar quantizer with fixed-length bit coding.

We identify a fundamental limitation of this approach: under the Gaussianized-coordinate model, any scalar fixed-length quantizer suffers an asymptotic distortion gap of at least $\pi\sqrt{3}/2 \approx 2.72$ relative to the Gaussian rate–distortion bound (Theorem 3). This gap decomposes into two independent penalties:

1. **Cell-shape penalty** ($\pi e/6 \approx 1.424$): from using 1-dimensional lattice cells instead of higher-dimensional ones.

2. **Fixed-length rate penalty** ($3\sqrt{3}/e \approx 1.911$): from assigning a fixed number of bits per sample instead of entropy coding.

We then establish a *general lattice achievability principle* (Theorem 5): for any n-dimensional lattice Λ, entropy-coded nearest-neighbor lattice quantization achieves a Gaussian-normalized distortion constant of $2\pi e\, G(\Lambda)$. This result applies to any continuous source with finite second moment and sufficiently regular density. The high-rate achievability extends to non-dithered nearest-

neighbor quantization (Lemma 4), which is the variant used in practice; compact integer storage of the quantized representation is a systems concern addressed in future work.

Among practical lattice choices, E_8 offers the strongest accuracy–complexity tradeoff: it achieves a gap of at most 1.224 (Corollary 6) while admitting efficient nearest-neighbor decoding (Conway–Sloane, 1982).

2 Related Work

Rotation-based quantization. QuaRot (Ashkboos et al., NeurIPS 2024) and SpinQuant (Liu et al., 2024) apply random or learned rotations to reduce outliers before scalar quantization. TurboQuant (Ashkboos et al., ICLR 2026) extends this with an online RHT and optimized Lloyd–Max scalar quantizer, achieving strong practical results for KV cache compression. All of these use scalar quantization with fixed-length coding, inheriting the 2.72× asymptotic barrier identified in Theorem 3.

Lattice-based quantization for LLMs. QuIP# (Tseng et al., ICML 2024) introduced lattice codebooks for weight quantization. NestQuant (Savkin et al., ICML 2025) implements E_8 quantization for weights, activations, and KV caches using a nested lattice structure with fixed-rate coding. Our framework differs in two respects: (1) we add entropy coding, removing the fixed-rate penalty that nested lattice schemes retain; (2) we provide the general achievability principle $2\pi e\, G(\Lambda)$ that explains *why* lattice quantization outperforms scalar approaches, rather than demonstrating a specific lattice implementation.

Information-theoretic foundations. The high-rate quantization theory underlying our results is classical (Gersho–Gray, 1992; Gray–Neuhoff, 1998; Zamir, 2014). The Panter–Dite formula (1951) for optimal companding and the Crypto Lemma (Zamir–Feder, 1996) for dithered lattice quantization are standard tools. Our contribution is applying these tools to derive sharp, practical bounds for the KV cache quantization problem.

3 Background and Setup

3.1 Source Model

Following TurboQuant, a key or value vector $\mathbf{v} \in \mathbb{R}^d$ is rotated by a randomized Hadamard transform: $\mathbf{w} = \frac{1}{\sqrt{d}} H D_\varepsilon \mathbf{v}$, with $\|\mathbf{w}\|^2 = \|\mathbf{v}\|^2 = S$ and per-coordinate variance $\sigma^2 = S/d$.

For the theoretical analysis, we work under the *Gaussianized-coordinate model*: as d grows, the RHT coordinates are approximately i.i.d. $\mathcal{N}(0, \sigma^2)$.

3.2 TurboQuant Pipeline

TurboQuant's quantization consists of three stages: (1) RHT rotation $\mathbf{v} \mapsto \mathbf{w}$; (2) scalar quantization of each coordinate with $N = 2^b$ levels; (3) fixed-length coding with exactly b bits per value.

3.3 Reference Benchmark

For an i.i.d. $\mathcal{N}(0, \sigma^2)$ source at rate b bits/sample, the Gaussian rate–distortion bound is: $D^*_{\text{Gauss}} = \sigma^2 \cdot 4^{-b}$. This serves as the normalization denominator throughout.

4 Theorem 1: Scalar Fixed-Length Quantization Barrier

4.1 Three Schemes and Their Gaps

Fix a Gaussian source $X \sim \mathcal{N}(0, \sigma^2)$ and rate b bits/sample.

Entropy-coded uniform scalar quantizer. Setting the operational rate $R = h(X) - \log \Delta = b \ln 2$:

$$D_{\mathrm{EC}} = \frac{\pi e}{6}\, \sigma^2 \cdot 4^{-b} \cdot (1 + o(1)), \qquad D_{\mathrm{EC}}/D^*_{\mathrm{Gauss}} \to \pi e/6 \approx 1.424.$$

Optimal companding + fixed-length coding. $N = 2^b$ levels, point density $\lambda(x) \propto f(x)^{1/3}$ (Panter–Dite, 1951):

$$D_{\mathrm{FL}} = \frac{\pi\sqrt{3}}{2}\, \sigma^2 \cdot 4^{-b} \cdot (1 + o(1)), \qquad D_{\mathrm{FL}}/D^*_{\mathrm{Gauss}} \to \pi\sqrt{3}/2 \approx 2.720.$$

Gap decomposition. $\frac{\pi\sqrt{3}}{2} = \frac{\pi e}{6} \times \frac{3\sqrt{3}}{e}$.

4.2 Derivations

Proposition 1 (Panter–Dite for Gaussian)**.** *$D_{\mathrm{FL}} = (\pi\sqrt{3}/2)\, \sigma^2 \cdot 4^{-b} \cdot (1 + o(1))$ as $b \to \infty$.*

Proof. $D = B^3/(12N^2)$ where $B = \int f^{1/3}\, dx$. For Gaussian: $B^3 = 6\pi\sqrt{3}\,\sigma^2$. With $N^2 = 4^b$: $D = (\pi\sqrt{3}/2)\, \sigma^2 \cdot 4^{-b}$. □

Remark (Generality)**.** *$D = B(f)^3/(12N^2)$ holds for any density f. The constant $\pi\sqrt{3}/2$ is Gaussian-specific; the cell/rate decomposition is general.*

Proposition 2. *$D_{\mathrm{EC}} = (\pi e/6)\, \sigma^2 \cdot 4^{-b} \cdot (1 + o(1))$ as $b \to \infty$.*

Proof. $G(\mathbb{Z}) = 1/12$. $D_{\mathrm{EC}} = G(\mathbb{Z}) \cdot \exp(2h(X)) \cdot 4^{-b} + o(4^{-b}) = (\pi e/6)\, \sigma^2 \cdot 4^{-b} + o(4^{-b})$ (Gersho–Gray, 1992, Ch. 6). □

4.3 The Scalar Barrier Theorem

Theorem 3 (Scalar Fixed-Length Barrier)**.** *Under the high-rate Gaussianized-coordinate model, any scalar fixed-length quantization scheme satisfies:*

$$\liminf_{b \to \infty} \frac{D}{D^*_{\mathrm{Gauss}}} \geq \frac{\pi\sqrt{3}}{2} \approx 2.72, \qquad \frac{\pi\sqrt{3}}{2} = \frac{\pi e}{6} \times \frac{3\sqrt{3}}{e}.$$

The lower bound is asymptotically tight.

Remark (Application to TurboQuant)**.** *Since TurboQuant uses scalar quantization with fixed-length coding on RHT-rotated coordinates, it belongs to this class.*

Proof. D_{FL} is the minimum high-rate distortion among scalar quantizers with 2^b fixed-length levels. $\liminf D/D^* \geq \lim D_{\mathrm{FL}}/D^* = \pi\sqrt{3}/2$. □

Table 1: Lattice quantization constants. $G(\Lambda)$ from Conway–Sloane (1988, Table 2.3).

Lattice	n	$G(\Lambda)$	$2\pi e\, G(\Lambda)$	**Decoding**
$\mathbb{Z}$ (scalar)	1	0.0833	1.424	trivial
D_4	4	0.0766	1.308	fast
E_8	8	0.0717	**1.224**	efficient
BW_{16}	16	0.0684	1.168	expensive
Λ_{24} (Leech)	24	0.0658	1.123	very expensive

5 Theorem 2: General Lattice Achievability

Lemma 4 (High-rate equivalence)**.** *For an n-dimensional lattice Λ scaled by $a > 0$ and a source with bounded, piecewise-smooth density, non-dithered nearest-neighbor quantization achieves, as $a \to 0$:* (i) $\frac{1}{n}\mathbb{E}[\|\hat{\mathbf{x}} - \mathbf{x}\|^2] = G(\Lambda) \cdot \mathrm{Vol}(\mathcal{V}_a)^{2/n} \cdot (1 + o(1))$*;* (ii) $\frac{1}{n}H(Q_{a\Lambda}(\mathbf{x})) = [h(\mathbf{x}) - \log \mathrm{Vol}(\mathcal{V}_a)]/n + o(1)$ (*Gersho–Gray, 1992; Gray–Neuhoff, 1998*)*.*

Theorem 5 (General Lattice Achievability)**.** *Let Λ be an n-dimensional lattice with $G(\Lambda)$, and $\mathbf{x} \in \mathbb{R}^n$ continuous with $\mathbb{E}[\mathbf{x}] = \mathbf{0}$, $\mathrm{tr}(\mathrm{Cov}(\mathbf{x})) = n\sigma^2$, and sufficiently regular density. Then entropy-coded nearest-neighbor Λ-quantization achieves:*

$$D \;\le\; G(\Lambda) \cdot 2\pi e \cdot \sigma^2 \cdot 4^{-b} \cdot (1 + o(1)) \qquad (b \to \infty), \tag{1}$$

at operational rate $\le b$ bits/dim (leading order). The bound does not require Gaussianity, independence, or a specific distributional form.

Proof sketch. The dithered quantizer achieves exact MSE $= G(\Lambda) \cdot \mathrm{Vol}(\mathcal{V}_a)^{2/n}$ (Crypto Lemma). The entropy bound $h(\mathbf{x}) \le (n/2)\log(2\pi e\sigma^2)$ (AM–GM + max-entropy) and the scale choice $\mathrm{Vol}(\mathcal{V}_a)^{2/n} = 2\pi e\sigma^2 \cdot 4^{-b}$ yield (1). Lemma 4 extends to nearest-neighbor, absorbed into $(1 + o(1))$. Full proof in Appendix A. □

Corollary 6 (E_8 Achievability)**.** *For $\Lambda = E_8$ ($n = 8$, $G(E_8) \approx 0.0717$): $D/D^*_{\mathrm{Gauss}} \le 1.224 \cdot (1 + o(1))$.*

6 Implementation

6.1 Parity-Aware Entropy Coding

$E_8 = D_8 \cup (D_8 + \frac{1}{2}\mathbf{1})$, where $D_8 = \{\mathbf{z} \in \mathbb{Z}^8 : \sum z_i \text{ even}\}$. The even-sum constraint means 8 coordinates have only 7 degrees of freedom. Our coding scheme: (1) coset bit (1 bit/block); (2) 7 free coordinates, entropy coded per-coordinate conditioned on coset; (3) $\lfloor z_8/2 \rfloor$ via entropy coding (parity of z_8 is free from the constraint). This saves 0.125 bits/dim vs. naive coding.

6.2 The Role of the RHT

Theorem 5 is stated under the Gaussianized-coordinate model, in which the RHT serves as a worst-case regularizer that ensures the source satisfies the regularity conditions. In practice, however, KV cache vectors after the learned attention projection already exhibit well-distributed coordinates, so

Table 2: RHT ablation: gap D/D^* at 4 bits/dim. RHT is unnecessary for well-conditioned sources but serves as a worst-case regularizer for structured adversarial inputs.

Distribution	**No RHT**	**With RHT**
i.i.d. Gaussian	1.224	1.224
Adversarial (2-dim energy)	0.097	0.717
Real KV (Llama-3.2-1B)	1.223	1.225

Table 3: Gaussian validation: MSE ratio (measured/theory) and ANS-coded rate ($N = 2 \times 10^6$, parity-aware coding). Theory gap = 1.2246.

b	MSE ratio	Gap	ANS rate	rate/b
2.0	1.000	1.224	2.075	1.037
2.5	1.000	1.224	2.538	1.015
3.0	1.000	1.224	3.020	1.007
4.0	1.000	1.224	4.005	1.001
5.0	1.000	1.224	5.002	1.000
6.0	1.000	1.224	6.001	1.000

the RHT is a theoretical safeguard rather than a practical necessity. Table 2 confirms this: the E_8 quantizer achieves the predicted gap *without* RHT on real KV caches.

The RHT serves as a worst-case guarantee: for sources with concentrated energy, it spreads energy across coordinates, restoring a more isotropic regime under which the quantizer behavior aligns with the theoretical Gaussianized-coordinate model. Note that the adversarial no-RHT gap of 0.097 is *artificially low*: when energy concentrates in two coordinates, the effective quantization problem collapses to two dimensions and the E_8 lattice's eight-dimensional structure is not exercised. The RHT restores the isotropic regime in which the Gaussian-normalized gap is meaningful. For well-conditioned sources—including real KV caches—the theoretical gap is achieved without RHT.

7 Experiments

Our experiments use the simplest possible implementation—per-head σ^2 estimation with no calibration data—to isolate the contribution of lattice geometry and entropy coding from orthogonal optimization techniques. Calibration-based refinements (LDLQ, per-channel scaling) are complementary and expected to further improve low-bitrate performance.

7.1 Gaussian Validation

We validate the E_8 quantizer on i.i.d. $\mathcal{N}(0, 1)$ data (10^6 blocks). Figure 1 shows the measured gap and Table 3 gives the numerical validation.

The measured distortion matches theory to within 0.05% across all bitrates. The ANS-coded bitstream achieves the target rate to within 1.5% at 2.5 bits/dim and $< 0.1\%$ at 5+ bits/dim. At 2

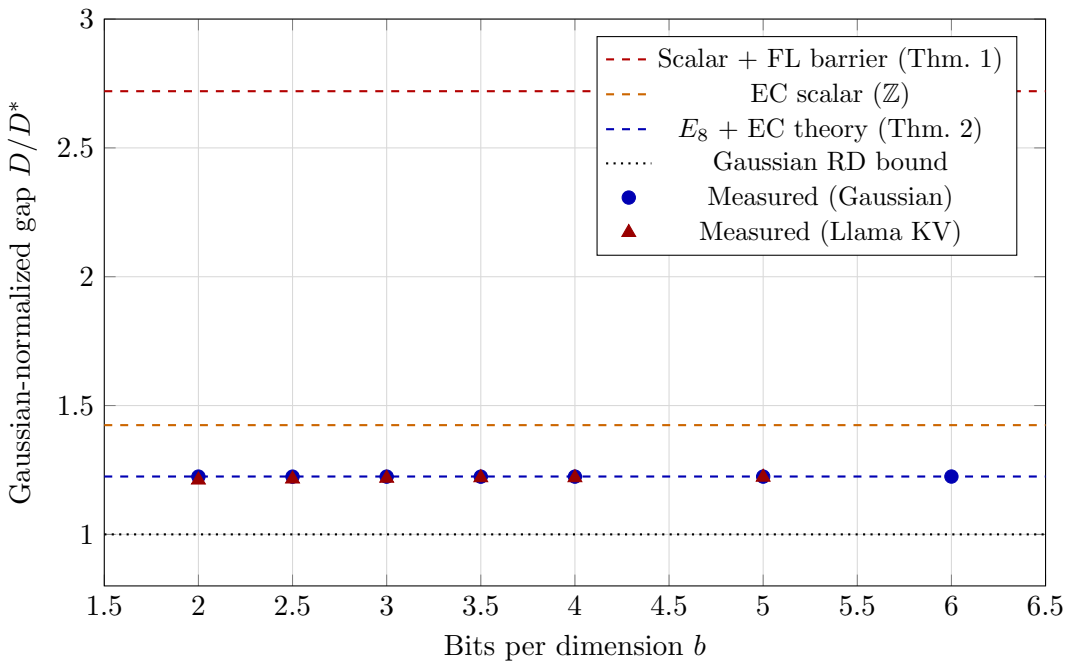

Figure 1: Gaussian-normalized distortion gap D/D^*. The E_8 quantizer tracks the theoretical prediction of 1.224, far below the scalar fixed-length barrier of 2.720.

bits/dim, the distortion prediction is already accurate, while the operational rate remains slightly above target, consistent with the asymptotic nature of the rate claim.

7.2 Real KV Cache Analysis

We apply the E_8 quantizer to KV cache tensors from Llama-3.2-1B (16 layers, 8 heads, head dim 64) with per-head σ^2 estimation.

The mean gap deviates from theory by only 0.34% at 4 bits/dim (Table 4). This shows that the Gaussian-normalized high-rate constant remains predictive even on real learned KV distributions.

7.3 End-to-End Perplexity

We evaluate perplexity on wikitext2 with quantized KV cache in the attention loop. The quantized cache is implemented via a custom `DynamicCache` subclass that applies E_8 nearest-neighbor quantization with per-head σ^2 estimation on every cache write. No calibration data is used. For Llama-3.1-8B, model weights are loaded in 8-bit precision via `BitsAndBytesConfig` to fit within 16 GB VRAM; KV cache quantization is applied independently. All reported perplexity changes are measured relative to the matched baseline (same weight precision, FP16 KV cache).

Two trends emerge from Table 5 and Figure 2. First, 4-bit quantization is practical for both model sizes: the perplexity increase is modest (+4.4% for 1B, +1.94% for 8B). Second, in our experiments, the larger model is more tolerant of quantization noise—the 8B model's degradation is approximately half that of the 1B model at every bitrate. These results suggest improved tolerance at larger scales, though confirming this trend requires evaluation on additional model sizes.

Table 4: KV cache quantization on Llama-3.2-1B. Theory gap = 1.2246.

b	Mean gap	Mean cos	Min cos
2.0	1.211	0.949	0.815
2.5	1.215	0.973	0.889
3.0	1.218	0.986	0.927
3.5	1.220	0.993	0.962
4.0	1.220	0.996	0.977
5.0	1.222	0.999	0.994

Table 5: Perplexity on wikitext2 with E_8 KV cache quantization. No calibration is used. For 1B, weights remain FP16; for 8B, weights are loaded in 8-bit precision via BitsAndBytes.

	Llama-3.2-1B			**Llama-3.1-8B**		
Config	**PPL**	**Δ**	**Compr.**	**PPL**	**Δ**	**Compr.**
FP16 baseline	8.62	—	1×	5.64	—	1×
5.0b E_8 + EC	8.74	+1.3%	3.2×	5.67	+0.51%	3.2×
4.0b E_8 + EC	9.00	+4.4%	4×	5.75	+1.94%	4×
3.5b E_8 + EC	9.44	+9.4%	4.6×	5.88	+4.23%	4.6×
3.0b E_8 + EC	10.89	+26.3%	5.3×	6.20	+9.77%	5.3×

7.4 Memory Savings Analysis

The practical motivation for KV cache quantization is memory reduction. Table 6 shows the theoretical KV cache sizes for representative configurations.

For a 70B-parameter model (80 layers, 8 KV heads, head dim 128), the theoretical KV cache footprint at 128K context is 40 GB under FP16—nearly filling an 80 GB A100. At 4 bits/dim, this reduces the theoretical KV-cache footprint to 10 GB, in principle freeing roughly 30 GB for batch serving or longer contexts. The memory formula is:

$$\text{KV cache} = 2 \times n_{\text{layers}} \times n_{\text{KVheads}} \times d_{\text{head}} \times L \times (b/8) \ \text{ bytes},$$

where L is the sequence length and b is the bits per dimension.

8 Discussion

The distortion constant $2\pi e\, G(\Lambda)$ depends on the lattice only through $G(\Lambda)$, separating lattice design from source coding. For E_8, the combined gains reduce the Gaussian-normalized distortion constant from at least 2.720 (scalar + fixed-length) to at most 1.224, corresponding to a 2.22× reduction, consistent with our empirical results.

Comparison with TurboQuant. TurboQuant inherits the 2.72× gap from scalar cells and fixed-length coding (Theorem 3). Its practical strength lies in system-level optimizations (online operation, attention-aware scaling). Our results are complementary: the lattice principle identifies *where* the distortion comes from and *how* to reduce it, while TurboQuant-style optimizations can be layered on top.

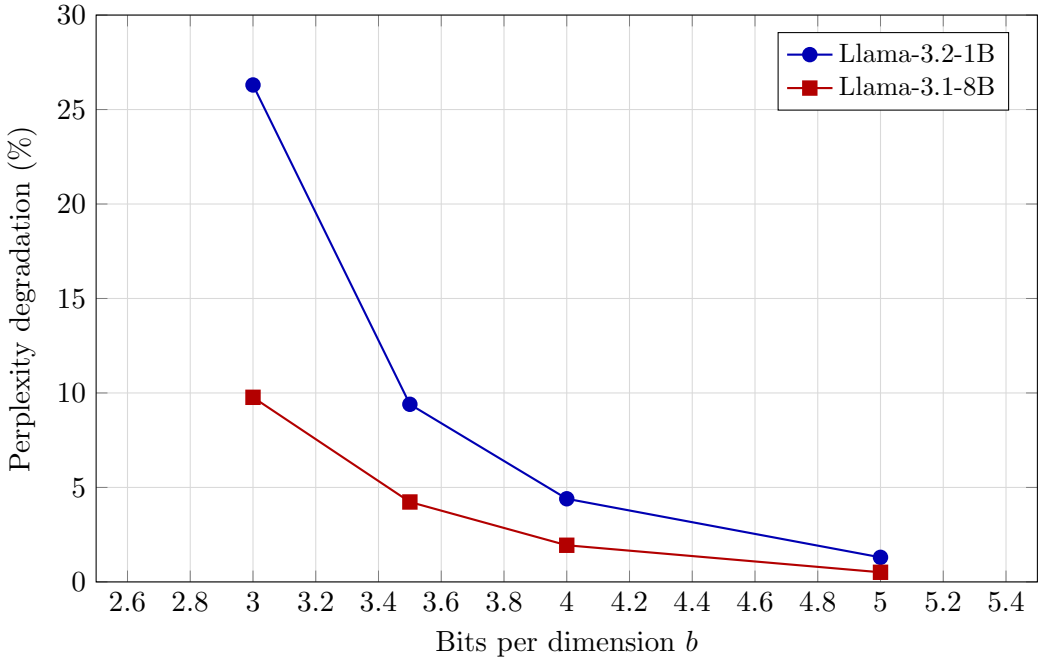

Figure 2: Perplexity degradation (% increase over FP16 baseline) on wikitext2 with quantized KV cache. In our experiments, the 8B model exhibits greater resilience to quantization than the 1B model: at 4 bits/dim, the 8B model degrades by only +1.94% versus +4.4% for the 1B model.

Comparison with NestQuant. NestQuant implements E_8 quantization via nested lattice structure with inherently fixed-rate coding. Our approach adds variable-rate entropy coding, which removes the $3\sqrt{3}/e \approx 1.911$ fixed-rate penalty. Additionally, we provide the general principle $2\pi e\, G(\Lambda)$ that applies to any lattice, not only E_8.

Scaling behavior. In our experiments, the 8B model is more tolerant of quantization noise than the 1B model: at 4 bits/dim, degradation is +1.94% versus +4.4%. This is consistent with the intuition that larger models have greater representational redundancy in their KV caches, though confirming this trend across a wider range of model sizes remains future work.

Limitations. Our experiments use a deliberately simple implementation to isolate the lattice geometry contribution. Calibration-based refinements (LDLQ, per-channel scaling, Hessian-aware optimization) are complementary and expected to further reduce perplexity degradation, particularly in the 3-bit regime. The current implementation simulates quantization quality within the standard `DynamicCache` interface; a production deployment would require storing quantized representations in compact integer format with on-the-fly dequantization, which we leave to future systems work. Decoding latency analysis of the entropy coding overhead is also left to future work, though we note that ANS decoding is $O(n)$ and GPU-parallel implementations exist (Weissenberger–Schmidt, 2019).

9 Conclusion

We have established that the scalar fixed-length quantization approach used by TurboQuant and related methods suffers an inherent Gaussian-normalized distortion constant of at least 2.72, decomposed into cell-shape and fixed-rate components. Our general lattice achievability theorem

Table 6: Theoretical KV cache storage footprint for Llama-3.1-8B (32 layers, 8 KV heads, head dim 128), assuming quantized values are stored in compact integer representation. At 128K context, the FP16 footprint exceeds the capacity of a single A10 or RTX 4090 (24 GB); 4-bit quantization reduces the theoretical footprint to 4 GB.

Context	**FP16**	**4-bit**	**3-bit**	**Savings**
4K	0.50 GB	0.12 GB	0.09 GB	4×
32K	4.0 GB	1.0 GB	0.75 GB	4×
128K	16.0 GB	4.0 GB	3.0 GB	4×
512K	64.0 GB	16.0 GB	12.0 GB	4×

shows that entropy-coded nearest-neighbor lattice quantization reduces this gap to $2\pi e\, G(\Lambda)$ for any lattice Λ. For E_8, this yields a gap of at most 1.224—a 2.22× reduction relative to the scalar barrier.

Experiments on Llama-3.2-1B and Llama-3.1-8B confirm the theory: the measured distortion gap on real KV caches deviates from prediction by less than 0.4%, and 4-bit quantization—corresponding to a 4× theoretical storage reduction—incurs only +1.94% perplexity increase on the 8B model. In our experiments, the larger model exhibits greater tolerance to quantization noise, suggesting that lattice-based KV quantization may become even more attractive at larger model scales. These results demonstrate that higher-dimensional lattice geometry and variable-rate coding are promising directions for next-generation KV cache compression, complementary to existing system-level optimizations.

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A Full Proof of Theorem 5

We prove the bound first for dithered quantization (where the MSE formula is exact), then invoke Lemma 4 to extend to nearest-neighbor.

Step 1: Distortion (exact, source-independent). Scale Λ by $a > 0$ and apply subtractive dithered quantization with shared dither $\mathbf{u} \sim \mathrm{Uniform}(\mathcal{V}_a)$: $\hat{\mathbf{x}} = Q_{a\Lambda}(\mathbf{x} + \mathbf{u}) - \mathbf{u}$. By the Crypto Lemma (Zamir–Feder, 1996, Thm. 2), $\mathbf{e} = \hat{\mathbf{x}} - \mathbf{x} \sim \mathrm{Uniform}(\mathcal{V}_a)$, independent of $\mathbf{x}$:

$$D(a) = \frac{1}{n}\mathbb{E}[\|\mathbf{e}\|^2] = G(\Lambda) \cdot \mathrm{Vol}(\mathcal{V}_a)^{2/n}. \tag{2}$$

Step 2: Entropy upper bound (exact). With eigenvalues $\lambda_1, \ldots, \lambda_n$ of $\mathrm{Cov}(\mathbf{x})$ summing to $n\sigma^2$:

$$\det(\mathrm{Cov}(\mathbf{x})) = \prod \lambda_i \le \sigma^{2n} \quad \text{(AM–GM)},$$

$$h(\mathbf{x}) \le \tfrac{1}{2}\log\det(2\pi e\,\Sigma) \le \frac{n}{2}\log(2\pi e\,\sigma^2) \quad \text{(max-entropy)}.$$

Step 3: Operational rate (high-rate).

$$R_{\mathrm{op}} = \frac{h(\mathbf{x}) - \log \mathrm{Vol}(\mathcal{V}_a)}{n} + o(1) \qquad (a \to 0).$$

(Zamir, 2014, Ch. 4; Gersho–Gray, 1992, Ch. 6.)

Step 4: Scale selection. Choose a so the worst-case source ($\mathcal{N}(\mathbf{0}, \sigma^2 I_n)$) uses rate b bits/dim: $\mathrm{Vol}(\mathcal{V}_a)^{2/n} = 2\pi e\,\sigma^2 \cdot 4^{-b}$. Substituting into (2): $D \le G(\Lambda) \cdot 2\pi e\,\sigma^2 \cdot 4^{-b}$.

Rate verification: for any source with $h(\mathbf{x}) \le h_{\max}$, $R_{\mathrm{op}} \le b \ln 2 + o(1)$.

Step 5: Aggregation. For $\mathbf{w} \in \mathbb{R}^d$ partitioned into $K = d/n$ blocks: $D_{\mathrm{total}} = (1/d)\sum_k \mathbb{E}[\|\mathbf{e}_k\|^2] = K \cdot nD/d = D$. (Linearity of expectation; dithered MSE depends only on scale.)

Step 6: Extension to nearest-neighbor. By Lemma 4, under the regularity conditions, non-dithered nearest-neighbor quantization achieves the same leading-order MSE and rate, absorbed into $(1 + o(1))$. □